

DPT Exercises

Exercise sheet 0

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Exercise 1. Let $\Omega \neq \emptyset$ and let $\{A_\alpha \in \mathcal{P}(\Omega) \mid \alpha \in I\}$ be an arbitrary set of subsets of Ω , I the indexing set. Then the following two identities hold (commonly referred to as *de Morgan's law*):

$$\left(\bigcap_{\alpha} A_{\alpha} \right)^c = \bigcup_{\alpha} A_{\alpha}^c, \quad \left(\bigcup_{\alpha} A_{\alpha} \right)^c = \bigcap_{\alpha} A_{\alpha}^c$$

Solution: We show the first statement by showing both inclusion " $\subseteq$ " and " $\supseteq$ ". The second statement follows analogously.

$\subseteq$: Let $x \in \left(\bigcap_{\alpha} A_{\alpha} \right)^c$. Then there exists an $\alpha \in I$ such that $x \notin A_{\alpha}$, meaning that $x \in A_{\alpha}^c$. Therefore $x \in \bigcup_{\alpha} A_{\alpha}^c$. As x is arbitrary, we have shown that $\left(\bigcap_{\alpha} A_{\alpha} \right)^c \subseteq \bigcup_{\alpha} A_{\alpha}^c$.

$\supseteq$: Let $x \in \bigcup_{\alpha} A_{\alpha}^c$. Then there exists an $\alpha \in I$ such that $x \in A_{\alpha}^c$ meaning that $x \notin A_{\alpha}$. Since $A_{\alpha} \subseteq \bigcap_{\beta} A_{\beta}$, we have that $x \notin \bigcap_{\beta} A_{\beta}$ and thus that $x \in \left(\bigcap_{\alpha} A_{\alpha} \right)^c$. Therefore, $\bigcup_{\alpha} A_{\alpha}^c \subseteq \left(\bigcap_{\alpha} A_{\alpha} \right)^c$.

Exercise 2. Let Ω_1, Ω_2 be non-empty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and $\mathcal{C} \subseteq \mathcal{P}(\Omega_2)$ an arbitrary collection of subsets of Ω_2 . Then the following statements hold.

(i) If $A, B \subseteq \Omega_2$ and $A \subseteq B$, then $f^{-1}(A) \subseteq f^{-1}(B)$.

(ii) The inverse image of the union is equal to the union of the inverse images, meaning

$$f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right) = \bigcup_{A \in \mathcal{C}} f^{-1}(A).$$

(iii) Given a subset $A \subseteq \Omega_2$, the inverse-image of the complement is equal to the complement of the inverse image, meaning

$$f^{-1}(A^c) = (f^{-1}(A))^c$$

Solution:

(i) Let $\omega \in f^{-1}(A)$. Then there exists $a \in A$ such that $f(\omega) = a$. Since $A \subseteq B$, we have $a = f(\omega) \in B$ and thus $\omega \in f^{-1}(B)$. Thus every element in $f^{-1}(A)$ is also in $f^{-1}(B)$.

(ii) Let $\omega \in f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right)$. Then there exists $A \in \mathcal{C}$ and $a \in A$ with $f(\omega) = a \in A$. Therefore, $\omega \in f^{-1}(A)$ and thus $\omega \in \bigcup_{A \in \mathcal{C}} f^{-1}(A)$. On the other hand let $\omega \in \bigcup_{A \in \mathcal{C}} f^{-1}(A)$. Then again, there exists an $A \in \mathcal{C}$ and $a \in A$ such that $f(\omega) = a$. Since $a \in A \subseteq \bigcup_{A \in \mathcal{C}} A$, we also have $\omega \in f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right)$.

(iii) Let $\omega \in f^{-1}(A^c)$. That means there exists an $a \in A^c$ such that $f(\omega) = a \in A^c$. Therefore $\omega \notin f^{-1}(A)$ and thus $\omega \in (f^{-1}(A))^c$. Vice versa, let $\omega \in (f^{-1}(A))^c$ then there exists an $a \in A^c$ such that $f(\omega) = a \in A^c$ and thus $\omega \in f^{-1}(A^c)$.

Exercise 3. Suppose $\mathcal{A}$ is a σ -algebra over $\Omega \neq \emptyset$, let $A_0, A_1, A_2, \dots$ be a countable collection of sets in $\mathcal{A}$ and let $m \in \mathbb{N}$. Then the sets

$$\Omega, \quad A_0 \setminus A_1, \quad \bigcup_{i=0}^m A_i, \quad \bigcap_{i=0}^m A_i, \quad \bigcap_{i \in \mathbb{N}} A_i$$

are also in $\mathcal{A}$.

[Hint: Use de Morgan's law for the intersection properties.]

Solution:

- (i) The entire set Ω is contained by an immediate consequence of property (ii), as

$$\Omega = \Omega \setminus \emptyset = \emptyset^c \in \mathcal{A}.$$

- (ii) For $A_0, A_1 \in \mathcal{A}$, the difference property immediately follows as

$$A_0 \setminus A_1 = A_0 \cap A_1^c \in \mathcal{A}.$$

- (iii) To see that the union of a finite collection $A_1, \dots, A_m \in \mathcal{A}$ is contained, note that the countable collection of sets $B_0, B_1, B_2, \dots$ is in $\mathcal{A}$, where

$$B_i := \begin{cases} A_i & i \leq m, \\ \emptyset & i > m. \end{cases}$$

We simply replaced the sets A_i for $i > m$ by the empty set (which is in $\mathcal{A}$ by (i)). Then the union effectively is only over the first m sets and the statement directly follows from property (iii) of a σ -algebra, i.e.,

$$A_1 \cup \dots \cup A_m = \bigcup_{i=1}^m A_i = \bigcup_{i \in \mathbb{N}} B_i \in \mathcal{A}.$$

- (iv) To show that the intersection of a finite collection $A_1, \dots, A_m \in \mathcal{A}$ is also in $\mathcal{A}$, we start by noting that by (i) of the σ -algebra properties the complements

$$A_1^c, \dots, A_m^c$$

are also in $\mathcal{A}$. By the finite union property (established above) we then have

$$\bigcup_{i=1}^m A_i^c \in \mathcal{A}$$

and by de-Morgans law and property (ii) of σ -algebras it follows that

$$\bigcap_{i=1}^m A_i = \left(\bigcup_{i=1}^m A_i^c \right)^c \in \mathcal{A}.$$

- (v) The intersection of an infinite collection of sets works analogously to the previous point.

Exercise 4. True or false? Justify your answers either by a proof or a counterexample.

- (i) The set $\mathcal{A} = \{\emptyset, \{1, 2\}, \{1, 4\}, \{2, 3\}, \{3, 4\}, \{1, 2, 3, 4\}\} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra over $\Omega = \{1, 2, 3, 4\}$.
(ii) If $\mathcal{A}_1, \mathcal{A}_2 \subseteq \mathcal{P}(\mathcal{A})$ are σ -algebras over Ω , then so is $\mathcal{A}_1 \cup \mathcal{A}_2$.
(iii) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is countable}\}$ is a σ -algebra over Ω .
(iv) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is finite}\}$ is a σ -algebra over Ω if and only if Ω is finite.

[Hint: You can use the following facts without further justification: The countable union of countable sets is again countable. Subsets of countable sets are again countable.]

Solution:

- (i) **False.** This can be seen, as the set collection does not satisfy the properties of a σ -algebra. For example, it does not satisfy the (finite) union property, as $\{1, 2, 3\} = \{1, 2\} \cup \{2, 3\}$ is not in $\mathcal{A}$, although $\{1, 2\}, \{2, 3\} \in \mathcal{A}$.
- (ii) **False.** A possible counter-example would be again the choice $\Omega = \{1, 2, 3, 4\}$ and σ -algebras $\mathcal{A}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$ and $\mathcal{A}_2 = \{\emptyset, \{1, 4\}, \{2, 3\}, \Omega\}$ (check that these are indeed σ -algebras). Then $\mathcal{A} := \mathcal{A}_1 \cup \mathcal{A}_2$ does not contain $\{1, 2, 3\}$ although $\{1, 2\}, \{2, 3\} \in \mathcal{A}$. Hence $\mathcal{A}_1 \cup \mathcal{A}_2$ is not a σ -algebra over Ω .
- (iii) **True.** We have to show that $\mathcal{A}$ satisfies the three properties of the definition of a σ -algebra.
- (a) $\emptyset \in \mathcal{A}$ because it is finite and thus countable.
- (b) Let $A \in \mathcal{A}$. Let us call a set whose complement is countable **co-countable**. Then we have to cases: (1) If A is countable, $A^c \in \mathcal{A}$, because A^c is co-countable. (2) If A itself is co-countable, then $A^c \in \mathcal{A}$, because A^c is countable.
- (c) Let $A_0, A_1, \dots$ be a countable collection of sets in $\mathcal{A}$. We have to distinguish between the following two cases:
- All sets A_i are countable. Then, by using the hint, $\bigcup_{i \in \mathbb{N}} A_i$ is countable as a countable union of countable sets.
 - There exists at least one $j \in \mathbb{N}$ such that A_j is uncountable and only its complement A_j^c is countable. Then by de Morgans law,

$$\left(\bigcup_{i \in \mathbb{N}} A_i \right)^c = \bigcap_{i \in \mathbb{N}} A_i^c \subseteq A_j^c.$$

Hence, the complement of the union is countable as the subset of a countable set. Therefore, the union $\bigcup_{i \in \mathbb{N}} A_i$ is co-countable, which concludes the proof.

- (iv) **True.** In order to establish the equivalence, we show both directions separately.

“ $\Rightarrow$ ”: Assume Ω is finite. Then it immediately holds that

$$\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is finite}\} = \mathcal{P}(\Omega),$$

which is always σ -algebra over Ω .

“ $\Leftarrow$ ”: We prove by contra-position, meaning we show that by means of a counter-example, that the set not being finite implies that $\mathcal{A}$ does not define a σ -algebra. Assume therefore $\Omega = \mathbb{N}$ is countably infinite. Then the sets contain only one element, $\{i\} \in \mathcal{A}$, but $2\mathbb{N} := \bigcup_{i \in \mathbb{N}} \{2i\} \notin \mathcal{A}$. Hence $\mathcal{A}$ is not a σ -algebra.

Exercise 5. Let $\Omega \neq \emptyset$ be a set, I be a nonempty index set (finite, countable or even uncountable) and let $\mathcal{A}_\alpha$ be a σ -algebra over Ω for each $\alpha \in I$ in the index set. Then $\mathcal{A} := \bigcap_{\alpha \in I} \mathcal{A}_\alpha$, the intersection of all the sigma-algebras is also a σ -algebra over Ω .

Solution: We have to show all three properties (i)-(iii) of a σ -algebra:

- (i) Since Ω is in $\mathcal{A}_\alpha$ for each $\alpha \in I$ (since $\mathcal{A}_\alpha$ σ -algebras) Ω is also in the intersection $\mathcal{A}$.
- (ii) Assume A is in the intersection $\mathcal{A}$. Then, by definition of the intersection, A is in $\mathcal{A}_\alpha$ for all $\alpha \in I$. This the $\mathcal{A}_\alpha$ are σ -algebras, $A^c \in \mathcal{A}_\alpha$ for all $\alpha \in I$, hence $A^c \in \mathcal{A}$.
- (iii) Let $A_1, A_2, \dots$ is in $\mathcal{A}$ be a countable collection of subsets in $\mathcal{A}$. Then $A_1, A_2, \dots$ are also in $\mathcal{A}_\alpha$ for each $\alpha \in I$. Since each $\mathcal{A}_\alpha$ is a σ -algebra, $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}_\alpha$ for each $\alpha \in \mathbb{N}$, hence $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}$.

Remark 1. Let Ω be a nonempty set and let $\mathcal{S} \subseteq \mathcal{P}(\Omega)$ also be nonempty. Then by Exercise 5, we have that

$$\mathcal{A}_\sigma(\mathcal{S}) := \bigcap \{ \mathcal{A} \subseteq \mathcal{P}(\Omega) \mid \mathcal{A} \subseteq \mathcal{S}, \mathcal{A} \text{ is a } \sigma\text{-algebra over } \Omega \}$$

is well-defined, a σ -algebra over Ω and by definition the smallest σ -algebra over Ω containing $\mathcal{S}$. We call $\mathcal{A}_\sigma(\mathcal{S})$ the σ -algebra generated by $\mathcal{S}$.

Exercise 6. Let Ω_1, Ω_2 be nonempty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and let $\mathcal{A}, \mathcal{B}$ be σ -algebras over Ω_1 and Ω_2 respectively. Then the *inverse image of $\mathcal{B}$ under f* and the *image or push-forward of $\mathcal{A}$ under f* defined as

$$f^{-1}(\mathcal{B}) := \{ f^{-1}(B) \mid B \in \mathcal{B} \} \quad \text{and} \quad f_*(\mathcal{A}) := \{ B \subseteq \Omega_2 \mid f^{-1}(B) \in \mathcal{A} \}$$

are σ -algebras over Ω_1 and Ω_2 respectively.

[Hint: Use statements from the second exercise.]

Solution:

- For $f_*(\mathcal{A})$, we have to again show the properties (i)-(iii) of σ -algebras:

(i) Since $\emptyset \in \mathcal{A}$ we have $\emptyset \in f_*(\mathcal{A})$, since $\emptyset = f^{-1}(\emptyset) \in \mathcal{A}$.

(ii) Let $B \in f_*(\mathcal{A})$. Then there exists a set $A \in \mathcal{A}$ such that $f^{-1}(B) = A$. Since $\mathcal{A}$ is a σ -algebra, $A^c \in \mathcal{A}$. By Exercise 2 (iii),

$$f^{-1}(B^c) = (f^{-1}(B))^c = A^c \in \mathcal{A},$$

and therefore $B^c \in f_*(\mathcal{A})$.

(iii) Suppose a collection of sets $B_1, B_2, \dots \in f_*(\mathcal{A})$. Then there exist sets $A_1, A_2, \dots \in \mathcal{A}$ such that $f^{-1}(B_j) = A_j$ for each $j \in \mathbb{N}$. Since $\mathcal{A}$ is a σ -algebra, $\bigcup A_j \in \mathcal{A}$. By Exercise 2 (ii),

$$f^{-1}\left(\bigcup_{j \in \mathbb{N}} B_j\right) = \bigcup_{j \in \mathbb{N}} f^{-1}(B_j) \in \mathcal{A},$$

therefore $\bigcup_{j \in \mathbb{N}} B_j \in f_*(\mathcal{A})$.

- The proof for $f^{-1}(\mathcal{B})$ works completely analogously.